Miter build

A miter merges spec and impl and XORs their outputs , equivalent designs yield miter PO=0 everywhere

The idea

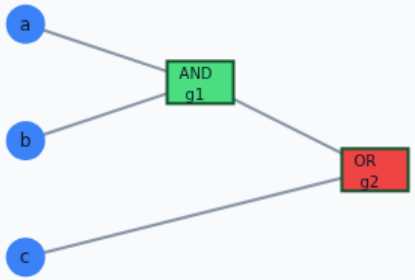
- Build_miter(spec, impl_good) adds i_-prefixed impl gates and miter_xor = spec_f XOR impl_f

XOR outputs

MITER

XOR outputs

Step 1 / 5 · idea · module01-03-miter-build



Explanation

A miter XORs the two POs under shared PIs.

- miter_out

```
diff detect
```

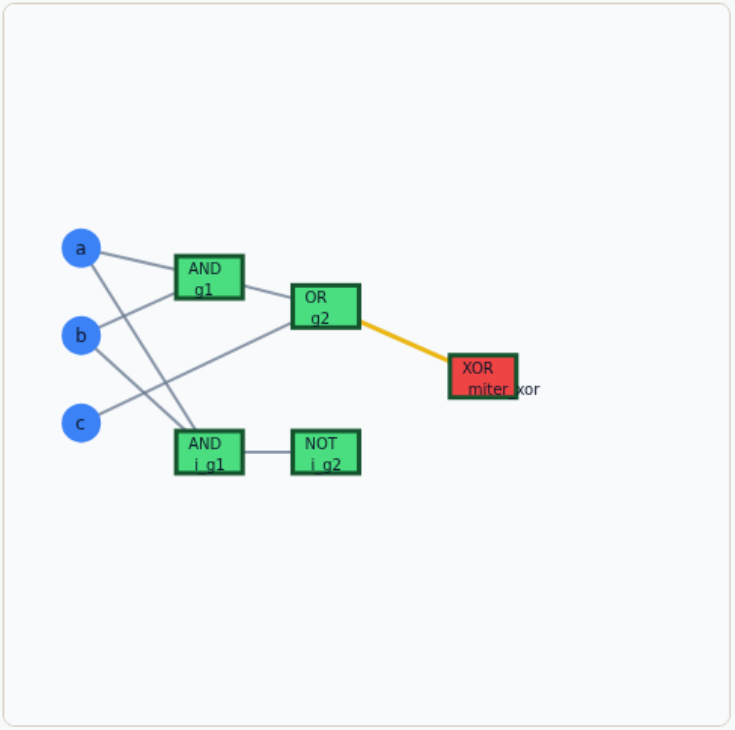
XOR outputs

Build good miter

MITER

Build good miter

Step 2 / 5 · build · module01-03-miter-build



Explanation

Prefix impl gates, XOR spec g2 with i_g5.

- build_miter

```
shared PIs
```

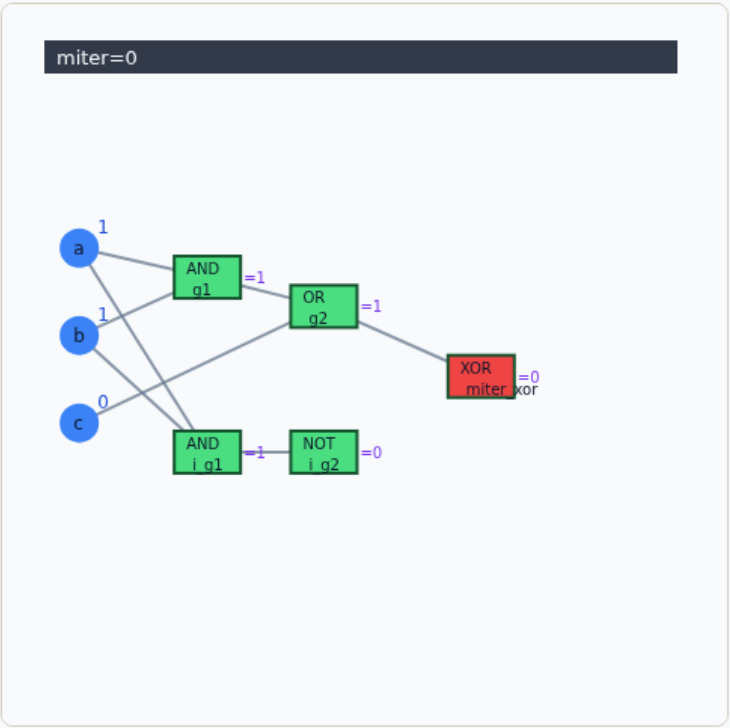
Build good miter

Good → all zero

MITER

Good → all zero

Step 3 / 5 · zero · module01-03-miter-build



Explanation

For the good pair, miter output is 0 on every input.

- equiv

```
8 zeros
```

Good → all zero

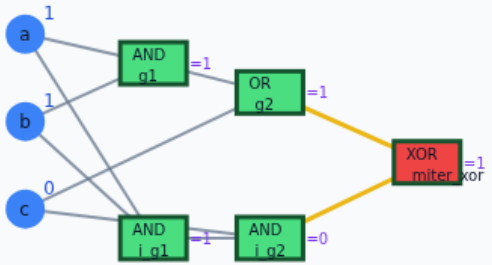
Bad → one

MITER

Bad → one

Step 4 / 5 · one · module01-03-miter-build

miter=1



Explanation

At the cex, bad miter outputs 1.

- diff

```
a1b1c0
```

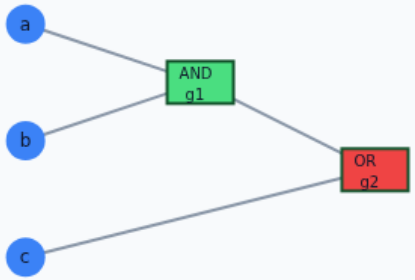
Bad → one

Why miters

MITER

Why miters

Step 5 / 5 · use · module01-03-miter-build



Explanation

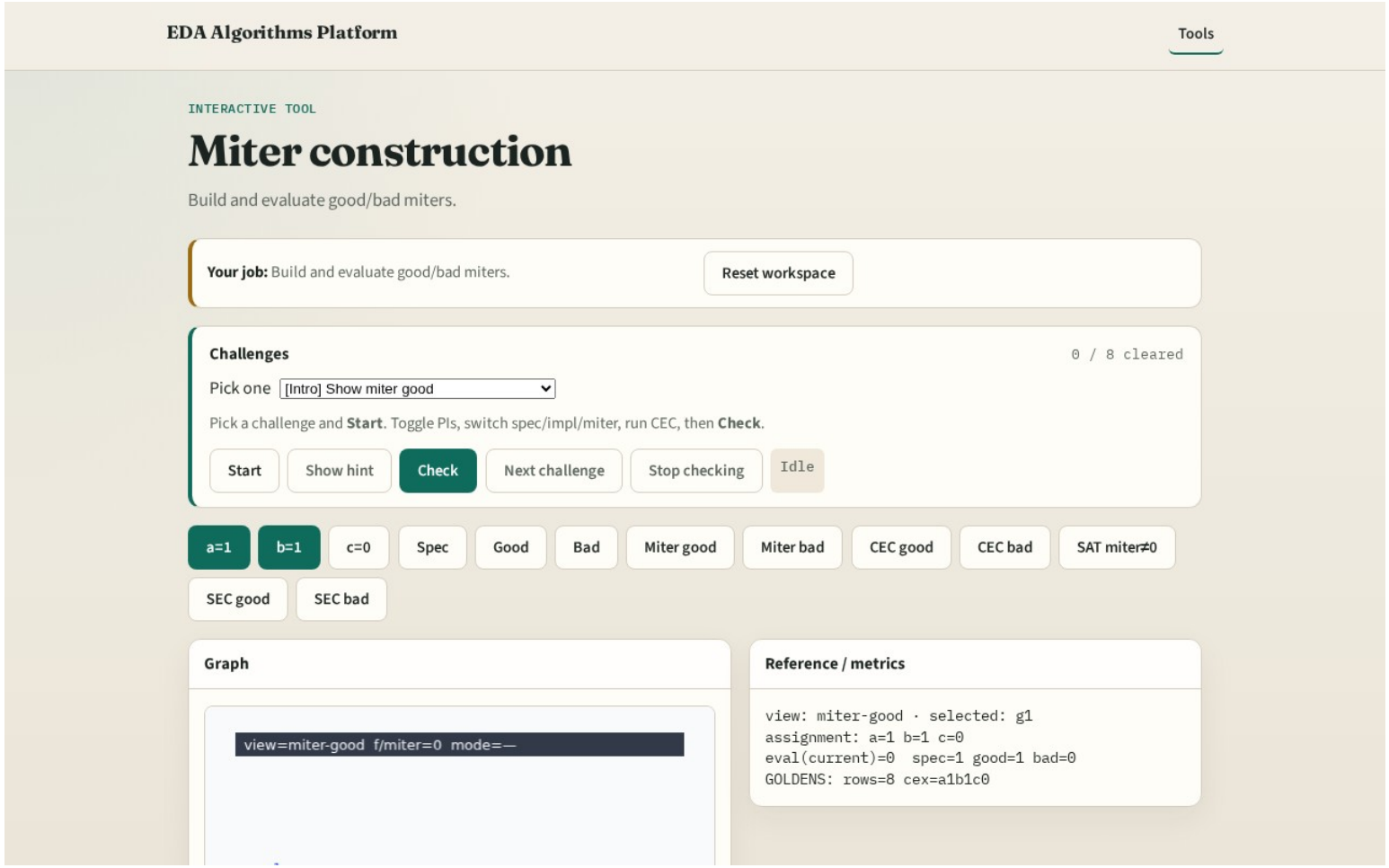
SAT/BDD CEC asks whether miter_out can be 1.

- UNSAT ⇒ equiv

```
Next TT
```

Why miters

Browser lab track



Browser lab starter

Implement track

- Call build_miter(spec, impl_good); miter_output must be 0 on every assignment

Pitfalls

- XORing inputs instead of outputs
- Forgetting to clear old output flags

Your turn

- Next: exhaustive truth-table CEC